Regional Shipping Time Analysis

Dataset: E-commerce orders · Generated: 2026-11-07 00:05 UTC

Question

Which region has the slowest shipping times, and is the difference between regions statistically significant?

Executive summary

The West region exhibits the slowest shipping times with an average of 4.20 days, which is higher than South (3.54 days), East (3.52 days), and North (3.49 days). Statistical testing confirms that the differences in shipping times across regions are highly significant ($p < 0.001$). Therefore, the West region has the highest average days to ship.

Scope & task alignment

Scope	Value
Metric	days_to_ship
Grouping	region
Task alignment	ALIGNED

Run metric	Value
iterations	2
tool calls	3
sql queries	1
findings verified	3/3
numeric claims supported	13/13
hallucination rate	0
task alignment	aligned
total tokens	19,616
latency s	57.17

Findings

1. West region has the highest average shipping time [VERIFIED]

Average days to ship for West is 4.2018 days, compared to South (3.5414), East (3.5171), and North (3.4913) as shown in the aggregated data.

Numbers checked: 4/4 matched tool outputs. Evidence: E1

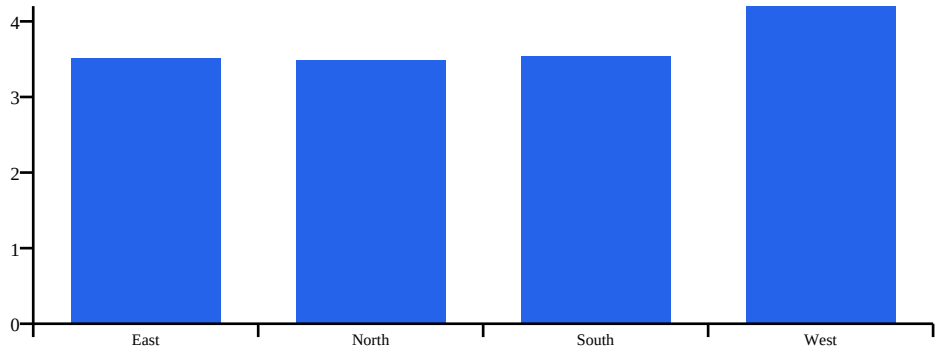
2. Shipping time differences across regions are statistically significant [VERIFIED]

One-way ANOVA yields a statistic of 414.55 with p■value 0.0, and Kruskal■Wallis H test gives 1125.04 with p■value 0.0, indicating significant variation in days_to_ship among regions.

Numbers checked: 4/4 matched tool outputs. Evidence: E2

Visualizations

Average Days to Ship by Region



Limitations & caveats

• Approximately 2.93% of days_to_ship values are missing, which could bias the averages.

• The analysis is observational; causality between region and shipping time cannot be established.

• Only aggregate statistics are provided; underlying distribution nuances are not explored.

Supporting evidence

[E0] profile_dataset - Dataset understanding

18861 rows, 13 columns; 0 warnings

[E1] run_sql - average days_to_ship per region

```
SELECT region, AVG(days_to_ship) AS avg_days_to_ship
FROM "ecommerce_orders"
GROUP BY region
ORDER BY avg_days_to_ship DESC;
```

region	avg_days_to_ship
West	4.2
South	3.54
East	3.52
North	3.49

[E2] compare_groups -

```
SELECT region, days_to_ship FROM "ecommerce_orders" WHERE days_to_ship IS NOT NULL
```

Differences in days_to_ship across region are statistically significant (p=0.0).

Agent execution timeline

#	Type	Step	Duration	Status
1	node	understand_dataset	97 ms	ok
2	tool	profile_dataset	97 ms	ok
3	discovery	mcp_tool_discovery	0 ms	ok
4	node	resolve_time_scope	43 ms	ok
5	node	plan	16581 ms	ok
6	llm	plan	16581 ms	ok
7	decision	task_contract	0 ms	ok
8	llm	act (iteration 1)	1172 ms	ok
9	node	reflect (iteration 1)	9367 ms	ok
10	llm	reflect	9367 ms	ok
11	decision	replan	0 ms	ok
12	llm	act (iteration 2)	2355 ms	ok
13	tool	run_sql	41 ms	ok
14	llm	act (iteration 2)	1596 ms	ok
15	tool	compare_groups	56 ms	ok
16	llm	act (iteration 2)	4367 ms	ok
17	tool	create_chart	61 ms	ok
18	llm	act (iteration 2)	1586 ms	ok
19	node	reflect (iteration 2)	7460 ms	ok
20	llm	reflect	7459 ms	ok
21	decision	stop_investigation	0 ms	ok

#	Type	Step	Duration	Status
22	node	synthesize	12321 ms	ok
23	llm	synthesize	12321 ms	ok
24	node	verify_findings	1 ms	ok
25	node	generate_report	0 ms	running